

RM operation SOPs

Scope

The grinding process starts with the preparation of the so-called ***the grinding bed*** where the feed is being crushed and then compressed and ground by the grinding rollers, in other words it can be said that the bed height is the cornerstone in the grinding process uses the vertical mills.

The ability of forming a stable grinding bed is essential for the mill stability, so a mix of both fine and coarse feed is required to form the grinding bed.

Too coarse materials can lead to a rough roller movement like a car on a gravel road while the too fine materials can lead to roller sliding or slipping like a car movement on a sandy road.

The optimum grinding bed height is 1 up to 2 % of the roller diameter.

The another parameter that widely influences the mill operation process is ***the mill pressure loss*** that depends on the amount of the dust inside mill and also on the gas flow through the mill, therefore it can be seen as the sum of the pressure loss resulting of the dust load plus the pressure loss of the air flow.

The dust load depends on material grindability, feed moisture, feed size, and also on grinding process efficiency.

The mill pressure loss equals $35 \sqrt{D_o}$, D_o is table diameter.

Mill operation conditions

1. Higher mill drive power accompanied by higher mill pressure loss

- Very high mill feed rates
- Separator over speed.
- Under draught
- Higher grinding pressure.

The Corrective actions to be as follow:

- Adjust mill feed rate
- Adjust the draught to the preset values.
- Adjust the separator to the preset values.
- Adjust the grinding pressure to the preset values.

2. Higher mill power accompanied by lower mill pressure loss

- Very low air flow.
- Under draught.
- Hard to grind materials.
- Coarse feed size.
- Higher water injection feed rates
- High grinding pressure values
- Worn out collecting scrapers

The Corrective actions to be as follow:

- Adjust the mill air flow.
- Use mix of fine and coarse mill feed.

- Adjust the grinding pressure to the preset values according to feed physical properties “size and HGI”.
- Adjust the water injection rates.

3. Lower mill power accompanied by higher mill pressure loss

- Very high air flow.
- Over draught.
- Easy to grind materials.
- Very fine feed.

The Corrective actions to be as follow:

- Adjust the mill air flow.
- Decrease the grinding pressure as less as possible.
- Start water injection.
- Use mix of fine and coarse mill feed.

4. Lower mill power accompanied by lower mill pressure loss

- Low mill feed rates
- Over draught.
- Very fine feed.

The Corrective actions to be as follow:

- Increase mill feed rate
- Adjust the mill air flow.
- Decrease the grinding pressure as less as possible.
- Start water injection.
- Use mix of fine and coarse mill feed.